Movement changes

If you take a Parkinson's drug that contains levodopa, such as co-beneldopa (Madopar) or co-careldopa (Sinemet), you may develop movement changes over time.

Find out about wearing off, "on/off" periods, and dyskinesia, and how they can be managed.

Key points

- Movement changes are most commonly linked to Parkinson's drugs that contain levodopa, including wearing off, "on/off" periods, and dyskinesia.
- Wearing off is when your medicine stops working before your next dose is due.
- "On/off" periods are changes in symptom control. You may go

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specialist or Parkinson's nurse. Treatment changes can often help manage them.

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Movement changes and levodopa

Levodopa can be very effective in controlling Parkinson's symptoms, especially in the early stages. Over time, as Parkinson's progresses, it may become less effective.

This can lead to:

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Some people may also develop involuntary movements called dyskinesia.

Wearing off

Wearing off happens when your Parkinson's medication does not last as long as it used to. This means your symptoms may start to come back before your next dose is due.

If this is affecting your day-to-day life, your specialist or Parkinson's nurse should review your medication.

In the early stages of Parkinson's, you may not notice wearing off. But as Parkinson's progresses, some people find the effects of their medication become less predictable.

Wearing off can affect more than movement. It can also affect other Parkinson's symptoms, including:

- anxiety
- sleepiness
- low mood
- pain.

"On/off" periods

"On" and "off" are terms used to describe how well your medication is working.

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start suddenly. Some people say it feels like a light switch being turned on and off.

Being “off” is not the same as freezing. Freezing affects specific movements rather than your whole body. For example, you may be unable to walk but still able to move your arms or speak.

There are different ways to manage “on/off” changes and freezing.

Read our information about freezing in Parkinson’s.

Dyskinesia

Dyskinesia is when you have involuntary movements you cannot control. These can include twitches, jerks, or twisting movements.

They can affect different parts of the body, including the arms, hands, legs, torso, face, or neck.

Dyskinesia affects people in different ways. Some people have movements for most of the day. Others only notice them after taking their medication, when levodopa levels are higher (sometimes called “peak dose dyskinesia”).

The movements can be mild or more severe. Mild movements may not bother you, even if others notice them. More severe dyskinesia can make everyday tasks, such as walking or talking, more difficult.

Because dyskinesia involves extra movement, some people may lose weight. If you’re worried about this, speak to your GP, specialist, or Parkinson’s nurse. They may review your medication or refer you to a dietitian.

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What causes motor fluctuations and dyskinesia?

Motor fluctuations are changes in how well symptoms are controlled. These changes can happen slowly or suddenly. They often happen when levodopa is wearing off, but can also happen at other times.

Many people experience motor fluctuations or dyskinesia after taking levodopa for a few years. Some people develop them earlier, especially if they are taking higher doses.

No one knows exactly why this happens. It's thought to be linked to changes in the brain caused by Parkinson's over time, and the way levodopa works in the brain.

The difference between dyskinesia and tremor

Dyskinesia and tremor are both involuntary movements, but they are different.

They can look similar and are sometimes confused. It's also possible to have both at the same time.

Key differences:

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the hand, it may look like a "pill-rolling" movement between the thumb and fingers. Dyskinesia tends to be more unsteady and unpredictable.

- Tremor often improves when you move the affected body part. Dyskinesia usually gets worse with movement.

If you're not sure which you're experiencing, speak to your specialist or Parkinson's nurse. It may help to record what happens and how it affects you.

Read our information about tremor and Parkinson's.

Managing movement changes

Movement changes can be challenging to manage.

Medication

If you lower your levodopa dose to reduce dyskinesia, your Parkinson's symptoms may become less well controlled. You may also have more frequent or longer periods of wearing off.

If you develop movement changes, talk to your specialist or Parkinson's nurse as soon as possible. They can review your medication and suggest changes to help balance symptom control and side effects.

They may suggest ways to reduce "peaks and troughs" in your medication. For example:

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symptoms stay more stable.

Apomorphine

Apomorphine may be recommended if you have severe “off” periods or dyskinesia that are not controlled by tablets alone.

Apomorphine is a dopamine agonist and does not contain levodopa. It’s a liquid medicine and can be given in two ways:

- as injections under the skin using a pen, when needed for “off” periods
- as a continuous infusion through a small portable pump, which gives a steady dose over the day.

It can be used on its own, but is often prescribed with levodopa. It’s not related to morphine used for pain relief.

Continuous medication treatments

Some people may be offered treatments that deliver Parkinson’s medication continuously. These are usually considered for people with advanced Parkinson’s.

Duodopa

Duodopa is a gel form of levodopa (co-careldopa). It’s given continuously through a small pump connected to a tube placed into the small intestine. The tube is fitted during a small operation.

This helps deliver medication in a steady way throughout the day.

Produodopa

Produodopa (foslevodopa-foscarbidopa) is a newer levodopa-

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vest.

Read our information about drug treatments for Parkinson's.

Deep brain stimulation

Deep brain stimulation (DBS) is a type of surgery that can help improve motor symptoms when medication is no longer working well, or if you have severe dyskinesia.

DBS can increase the amount of time you are "on". This may allow you to reduce your medication and side effects.

It's not suitable for everyone. Like all surgery, DBS has risks. Your specialist can talk to you about whether it may be suitable.

Read our information about deep brain stimulation for Parkinson's.

Diet

For some people, protein can affect how well levodopa is absorbed. Protein is found in foods such as meat, fish, eggs, beans, and lentils.

A small number of people with Parkinson's notice that protein in their diet can affect how well levodopa works, especially if they have motor fluctuations.

If this happens, speak to your specialist or GP.

You may be advised to:

- take medication 30 to 60 minutes before meals
- try eating most daily protein in the final main meal of the day (a protein redistribution diet).

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affect how well medication is absorbed. It can be caused by Parkinson's symptoms, lower activity levels, diet, or medicines. Some other medications, such as iron supplements, can also affect how well your Parkinson's medication is absorbed.

If you're struggling with constipation, speak to your specialist or Parkinson's nurse. They may refer you to a dietitian. Your pharmacist can also help if you have concerns about medicines or possible interactions.

- **Read our information about diet and Parkinson's.**
- **Read our information about bladder and bowel changes in Parkinson's.**

Managing stress

Stress can make movement changes worse.

It may help to try ways to relax, such as:

- breathing exercises
- physical activity or exercise (such as tai chi or yoga)
- massage.

If you often feel stressed or anxious, speak to your GP, specialist, or Parkinson's nurse. They may be able to refer you for talking therapies or other support.

- **Read our information about physical activity**
- **Read our information about complementary therapies**
- **Read our information about anxiety** .

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more “off” time with other Parkinson’s symptoms.

Many people prefer more “on” time, even with some dyskinesia. But this is a personal choice.

Speak to your specialist or Parkinson’s nurse about what is right for you.

Keeping a diary

Keeping a diary can help you and your health professional understand your symptoms and how your medication is working.

You or someone supporting you can record:

- when symptoms and changes happen
- how long they last
- when you take your medication and the dose
- any new symptoms, such as sleep changes, speech or swallowing problems, or confusion.

Some people who have periods notice that their motor symptoms are affected by their menstrual cycle. Research on how the menstrual cycle affects Parkinson’s symptoms is limited, but keeping a diary may help you spot patterns.

Your diary can help your health professional adjust your treatment if needed.

- **Read our information about monitoring your Parkinson’s.**
- **Read our information about women and Parkinson’s.**

Our helpline

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Get this information in other formats

If you prefer to have something printed to read or want to download a PDF, choose an option below:

- **Download our Movement changes in Parkinson's booklet (PDF, 223KB)**
- **Order a free printed booklet by phone or email**

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Read more about wearing off

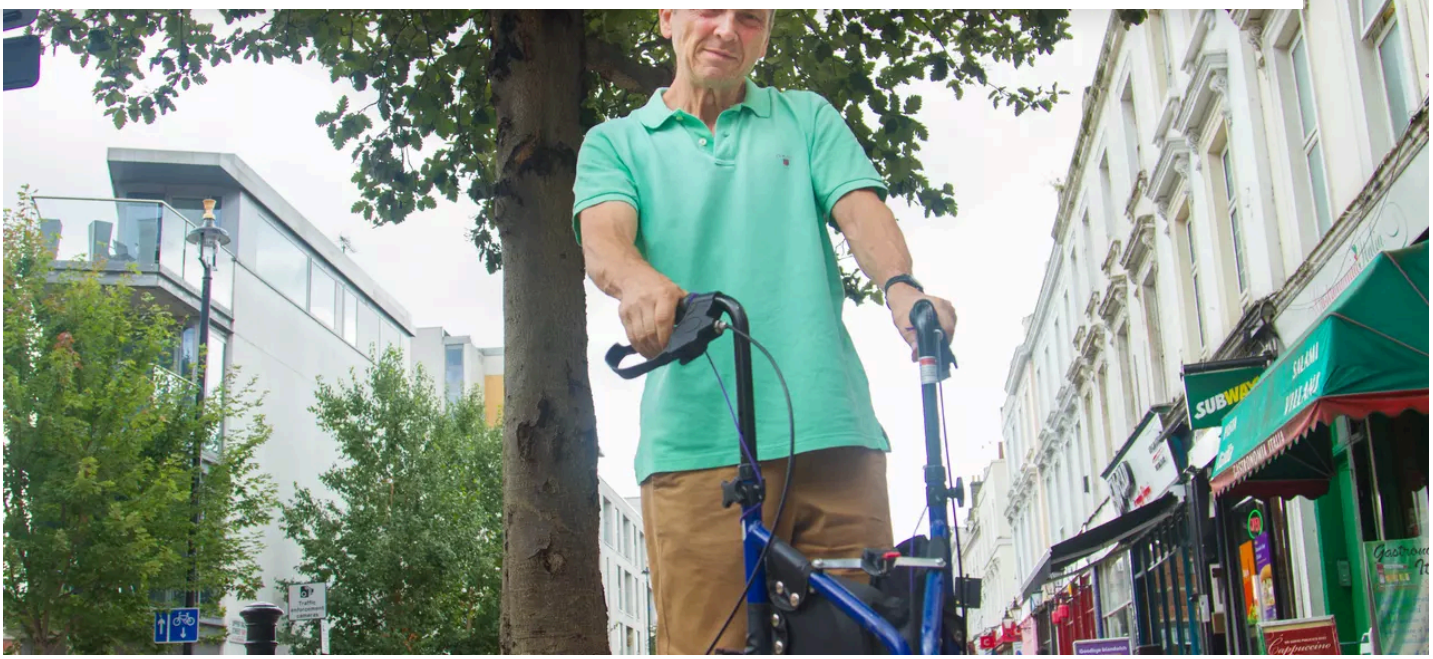


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Living with wearing off and dyskinesia

People with Parkinson's share their experience of living with wearing off and dyskinesia and give tips on how to manage these side effects of their...

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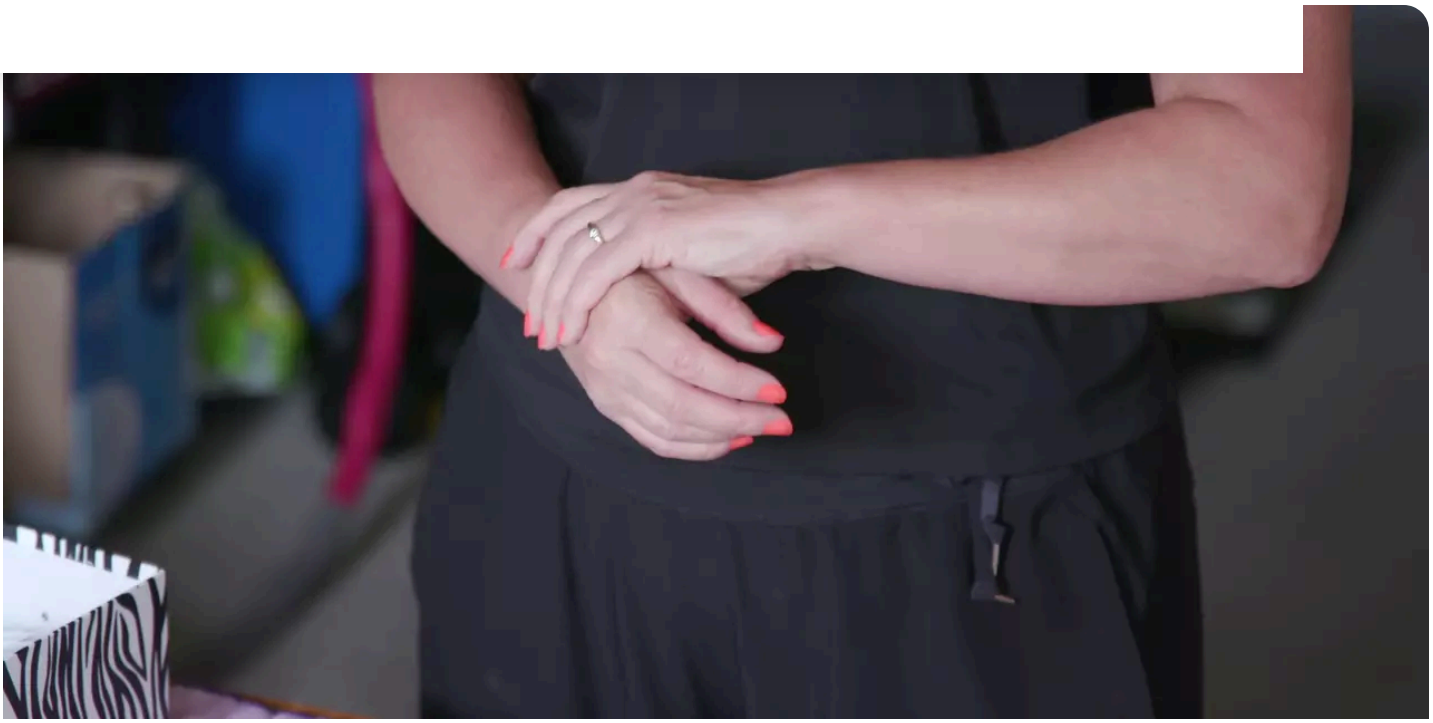


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Managing wearing off and dyskinesia with Professor Richard Walker

Wearing off and involuntary movements (dyskinesia) are movement problems that can affect people with Parkinson's. Below, Richard Walker, Consultant...

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Managing Parkinson's motor symptoms

Motor symptoms affect your movement. Specialist physiotherapist Bhanu Ramaswamy explains more about 3 common motor symptoms - tremor, dyskinesia...

Showing 3 of 3 articles

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Coping with dyskinesia

"It's almost as if I've forgotten how to walk," says Bob. "I go from day to day trying to change different things with my walking to improve it.

"I've been using this walker for about a year now and it's really made people want to help me more."

Watch our short film to hear from Bob about his experience of using a walker.

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Living with Parkinson's and wearing off – Paula's story

Paula shares how her Parkinson's symptoms impact her everyday life, and how 'wearing off' affects her.

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Thank you

Thank you to the health professionals that reviewed this information:

- Dr Duncan McLauchlan, Consultant Neurologist, Cardiff and Vale University Health Board
- Gina M Robinson, Programme Lead and Senior Lecturer in Nursing, University of Sunderland

Thanks also to our information review group and other people affected by Parkinson's who provided feedback. **Read more about joining our information review group.**

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